

● EASY

● ALGEBRA

● ANSWER KEY

Linear equations in two variables

02

10 answers · Rationale-rich · Teach from doc

Q1**C****C. 105**

Choice C is correct. It's given that the equation $x + y = 350$ relates the total number of maple trees, x , and the total number of birch trees, y , planted in a 14-acre forest preserve. It's also given that 245 maple trees were planted in the forest preserve. Substituting 245 for x in the given equation yields $245 + y = 350$. Subtracting 245 from both sides of this equation yields $y = 105$. Therefore, 105 birch trees were planted in the forest preserve.

Choice A is incorrect. This is the number of acres in the forest preserve, not the number of birch trees planted in the forest preserve.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the number of maple trees planted in the forest preserve, not the number of birch trees planted in the forest preserve.

Q2

D

D.

x	y
0	-5
1	-3
2	-1

Choice D is correct. It's given that the graph shows the linear relationship between x and y . The given graph passes through the points 0, -5, 1, -3, and 2, -1. It follows that when $x = 0$, the corresponding value of y is -5, when $x = 1$, the corresponding value of y is -3, and when $x = 2$, the corresponding value of y is -1. Of the given choices, only the table in choice D gives these three values of x and their corresponding values of y for the relationship shown in the graph.

Choice A is incorrect. This table represents a relationship between x and y such that the graph passes through the points 0, 0, 1, -7, and 2, -9.

Choice B is incorrect. This table represents a relationship between x and y such that the graph passes through the points 0, 0, 1, -3, and 2, -1.

Choice C is incorrect. This table represents a linear relationship between x and y such that the graph passes through the points 0, -5, 1, -7, and 2, -9.

Q3**B**

B. $0.02x + 0.96y = 0.0424$

Choice B is correct. It's given that a chemist mixes x kilograms of a low-salt mixture, which is 2% salt by weight. Multiplying 0.02 by the amount of the low-salt mixture, x kilograms, yields $0.02x$ kilograms of salt in the low-salt mixture. It's also given that the chemist mixes y kilograms of a high-salt mixture, which is 96% salt by weight. Multiplying 0.96 by the amount of the high-salt mixture, y kilograms, yields $0.96y$ kilograms of salt in the high-salt mixture. Therefore, the total amount of salt in the combined mixture is $0.02x + 0.96y$ kilograms. It's given that the low-salt mixture and the high-salt mixture together create 24 kilograms of a combined mixture that is 4% salt by weight. Thus, the amount of salt in the combined mixture is $0.04(24)$ kilograms. Since the total amount of salt in the combined mixture equals the amount of salt in the low-salt mixture and the amount of salt in the high-salt mixture, the equation $0.02x + 0.96y = (0.04)(24)$ represents this situation.

Choice A is incorrect. This equation represents a situation where the low-salt mixture is 96%, not 2%, salt by weight and the high-salt mixture is 2%, not 96%, salt by weight.

Choice C is incorrect. This equation represents a situation where the low-salt mixture is 96%, not 2%, salt by weight and the high-salt mixture is 2%, not 96%, salt by weight, and where the combined mixture contains 24 kilograms of salt, not 24 kilograms of a mixture that is 4% salt by weight.

Choice D is incorrect. This equation represents a situation where the combined mixture contains 24 kilograms of salt, not 24 kilograms of a mixture that is 4% salt by weight.

Q4**13**

The correct answer is 13. It's given that the equation $8x + 11y = 170$ gives the possible combinations of the number of 2009 premium grade Log Cabin Pennies, x , and the number of 1996 select grade Lincoln Pennies, y , in a collection that is worth a total of \$ 170. It's also given that there are 6 1996 select grade Lincoln Pennies in the collection, and thus $y = 6$. Substituting 6 for y in the given equation yields $8x + 11(6) = 170$, or $8x + 66 = 170$. Subtracting 66 from both sides of this equation yields $8x = 104$. Dividing both sides of this equation by 8 yields $x = 13$. Therefore, there are 13 2009 premium grade Log Cabin Pennies in the collection.

Q5**B****B. 4**

Choice B is correct. It's given that the equation $40x + 20y = 160$ represents the number of sweaters, x , and the number of shirts, y , that Yesenia purchased for \$ 160. If Yesenia purchased 2 sweaters, the number of shirts she purchased can be calculated by substituting 2 for x in the given equation, which yields $40(2) + 20y = 160$, or $80 + 20y = 160$. Subtracting 80 from both sides of this equation yields $20y = 80$. Dividing both sides of this equation by 20 yields $y = 4$. Therefore, if Yesenia purchased 2 sweaters, she purchased 4 shirts.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the number of shirts Yesenia purchased if she purchased 0 sweaters.

Choice D is incorrect. This is the price, in dollars, for each sweater, not the number of shirts Yesenia purchased.

Q6**D****D. 4**

Choice D is correct. Let p represent the number of pounds of potatoes and let c represent the number of pounds of celery that Davio bought. It's given that potatoes cost \$ 0.69 per pound and celery costs \$ 0.99 per pound. If Davio spent \$ 5.34 in total, then the equation $0.69p + 0.99c = 5.34$ represents this situation. It's also given that Davio bought twice as many pounds of celery as pounds of potatoes; therefore, $c = 2p$. Substituting $2p$ for c in the equation $0.69p + 0.99c = 5.34$ yields $0.69p + 0.99(2p) = 5.34$, which is equivalent to $0.69p + 1.98p = 5.34$, or $2.67p = 5.34$. Dividing both sides of this equation by 2.67 yields $p = 2$. Substituting 2 for p in the equation $c = 2p$ yields $c = 4$. Therefore, Davio bought 4 pounds of celery.

Choice A is incorrect. This is the number of pounds of potatoes, not the number of pounds of celery, Davio bought.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q7**B****B.** 770

Choice B is correct. It's given that the equation $x + y = 1,440$ represents the number of minutes of daylight, x , and the number of minutes of non-daylight, y , on a particular day in Oak Park, Illinois. It's also given that this day has 670 minutes of daylight. Substituting 670 for x in the equation $x + y = 1,440$ yields $670 + y = 1,440$. Subtracting 670 from both sides of this equation yields $y = 770$. Therefore, this day has 770 minutes of non-daylight.

Choice A is incorrect. This is the number of minutes of daylight, not non-daylight, on this day.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the total number of minutes of daylight and non-daylight.

Q8**A****A.** $-\frac{1}{3}$

Choice A is correct. It's given that line j is perpendicular to line k in the xy -plane. It follows that the slope of line j is the opposite reciprocal of the slope of line k . The equation for line k is written in slope-intercept form $y = mx + b$, where m is the slope of the line and b is the y -coordinate of the y -intercept of the line. It follows that the slope of line k is 3. The opposite reciprocal of a number is -1 divided by the number. Thus, the opposite reciprocal of 3 is $-\frac{1}{3}$. Therefore, the slope of line j is $-\frac{1}{3}$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q9**15**

The correct answer is 15. It's given that the equation $46 = 2x + 2y$ gives the perimeter of a rectangular rug that has length x , in feet, and width y , in feet. It's also given that the width of the rug is 8 feet. Substituting 8 for y in the equation $46 = 2x + 2y$ yields $46 = 2x + 28$, or $46 = 2x + 16$. Subtracting 16 from both sides of this equation yields $30 = 2x$. Dividing both sides of this equation by 2 yields $15 = x$. Since x represents the length, in feet, of the rug, it follows that the length of the rug is 15 feet.

Q10**D**

D. $y = 4x + 6$

Choice D is correct. A line in the xy -plane with a slope of m and a y -intercept of $0, b$ can be defined by an equation in the form $y = mx + b$. It's given that line r has a slope of 4 and passes through the point $0, 6$. It follows that $m = 4$ and $b = 6$. Substituting 4 for m and 6 for b in the equation $y = mx + b$ yields $y = 4x + 6$. Therefore, the equation $y = 4x + 6$ defines line r .

Choice A is incorrect. This equation defines a line that has a slope of -6, not 4, and passes through the point $0, 4$, not $0, 6$.

Choice B is incorrect. This equation defines a line that has a slope of 6, not 4, and passes through the point $0, 4$, not $0, 6$.

Choice C is incorrect. This equation defines a line that passes through the point $0, -6$, not $0, 6$.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

Generated 2026-05-29 · cb-educator-question-bank